

Miles H. Currie, Ph.D.

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Skills Summary

Coding	📖 Python (10+ years exp.), cluster computing, $\text{\LaTeX}$, FORTRAN, C/C++, Julia
Research	📖 Imaging/hyperspectral analysis, computer vision, data calibration pipeline development, machine learning for image analysis, Bayesian statistical inference, radiative transfer, gas abundance retrieval, time-series analysis, object perimeter detection and prediction, synthetic data generation, DIRSIG
Communication	📖 Strong reading, writing, and speaking competencies demonstrated through high-impact journal publications, conference proceedings, and invited seminars.

Education

2018 – 2024	📖 Ph.D. University of Washington, Seattle , Astronomy & Astrobiology (dual-title) Thesis: <i>Searching for Signs of Habitability and Life in the Era of Extremely Large Telescopes</i>
2018 – 2020	📖 M.Sc. University of Washington, Seattle , Astronomy
2013 – 2017	📖 B.Sc. Florida State University , Physics & Astrophysics, <i>magna cum laude</i>

Experience

2025 – present	📖 <i>Satellite AI/ML Scientist</i> , MyRadar. Satellite ML/AI edge computing: Leads machine learning development for hazard alerting platforms onboard satellites with multi/hyperspectral imaging, including for interplanetary space weather, wildfires, volcanoes, and nuclear fallout.
2024 – 2025	📖 <i>NASA Research Fellow</i> , NASA Goddard Space Flight Center. Imaging software development: Led a novel Python package for simulating observations with NASA's next flagship mission designed to detect and characterize objects 10^{10} times fainter than the background in the UV/VIS/NIR. Computer vision: Used computer vision techniques to characterize object morphology/composition and detect sub-pixel changes in IR imaging data over time. Object/pattern recognition: Employed Bayesian statistical analysis to build fingerprinting algorithms for UV/VIS/NIR imaging/spectroscopy, upgrading statistical frameworks for realism without compromising runtime. Data calibration: Developed custom data calibration improvements to increase sensitivity to a particular signal by up to a factor of ~ 2 .
2018–2024	📖 <i>Graduate Research Assistant</i> , Virtual Planetary Laboratory at UW Seattle. Low SNR image detection: Developed a statistical algorithm to detect spectroscopic features for objects 10^8 times fainter than the background in the VIS/IR. Image/data simulation and calibration: Built a Python package for simulating high spectral resolution ($R=100k$) data for future observatories and designed a foundational data calibration pipeline for these observations.
2017–2018	📖 <i>Post-baccalaureate Researcher</i> , Space Telescope Science Institute. Anomaly detection, spatial: Developed a novel convolutional neural network for automated anomaly detection, trained to handle undersampled time-series data. Data/image calibration: Developed a Bayesian mixture model for automated bad pixel detection and rejection for large imaging datasets that identified unstable pixels with better purity and completeness than previous methods.

Experience (continued)

Summer 2017, 2018

- *Wildfire Dynamics Researcher*, Geophysical Fluid Dynamics Institute.
Object perimeter detection and prediction: Developed a pixel-by-pixel wildfire spread and prediction model in Python, improving over previous models by employing spread probabilities derived from empirical field data.
Imaging field work: Deployed field sensors and ran experiments at prescribed burn sites, developing computer vision techniques for wildfire perimeter estimation.
Augmented reality imaging model: Developed an augmented reality system for fire spread analysis over changing topography by projecting real-time, computationally efficient models onto a physical sandbox.

2013–2017

- *Undergraduate Researcher*, Florida State University, NASA Ames Research Center, SETI Institute.
Data calibration: Developed a novel algorithm for cleaning unwanted systematic and natural noise from time-series data, based on a statistical principal component analysis of the data.

Publication Summary

- Journal Publications: **10** refereed + **3** non-refereed
- Conference Proceedings: **24**
- [Click for Link to Google Scholar Page](#)
- h-index: **8**; i10 index: **7**; total citations: **276**

Awards and Achievements

- 2024 ■ **Prize Fellowship**, NASA Goddard Space Flight Center
- 2023 ■ **Department Prize for Outstanding Graduate Student Research**, University of Washington.